

EE6347 - Assignment 4

Name: Himanshu Rajnish Borkar

Date: 26/10/2025

Roll No. : EE22B070

Task 1:

(a) Create 3 x 3 RRAM Crossbar using the same model used in the Mid-Sem exam

Ans:

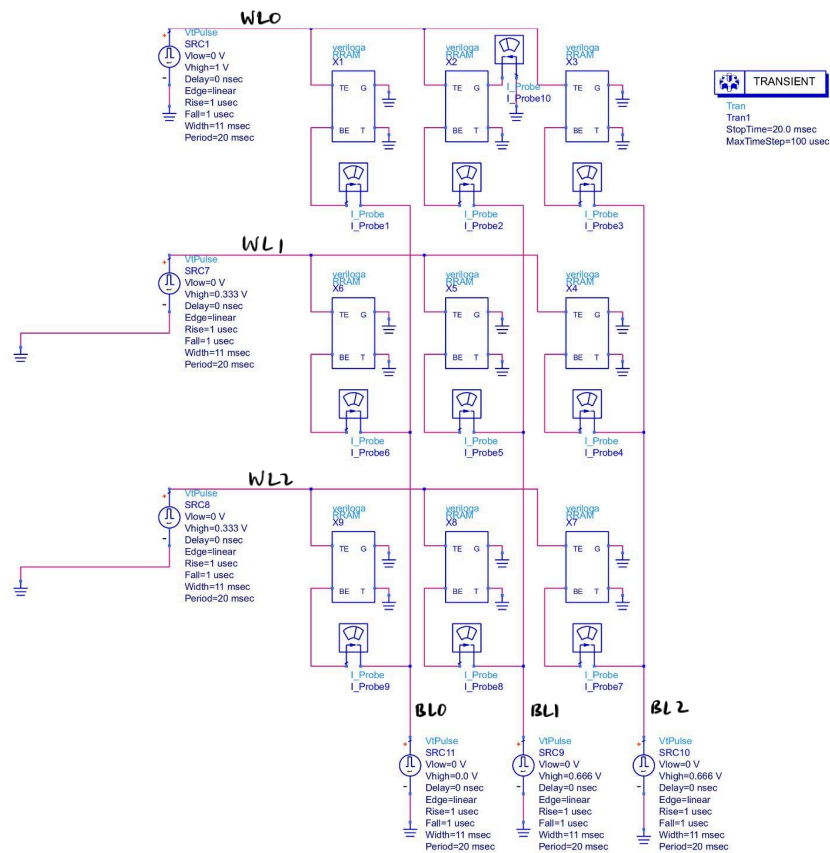


Figure 1. 3x3 RRAM crossbar schematic

(b) Follow the steps we followed in Tutorial 5. Use a V/3 scheme. Write the (0,0) cell and read currents through (1,0) and (2,2)

Ans:

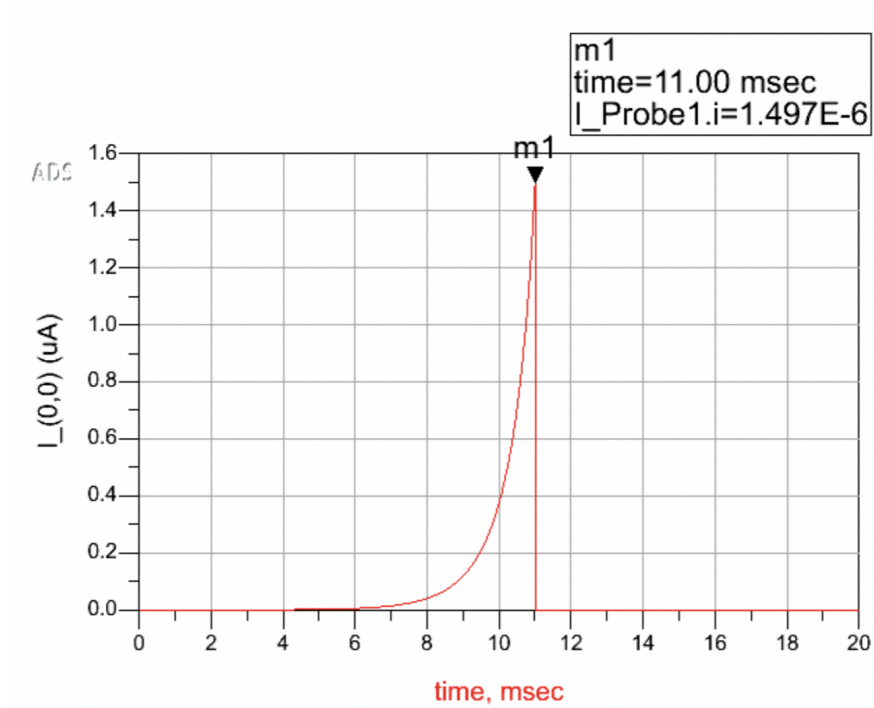


Figure 2. Current through cell(0,0)

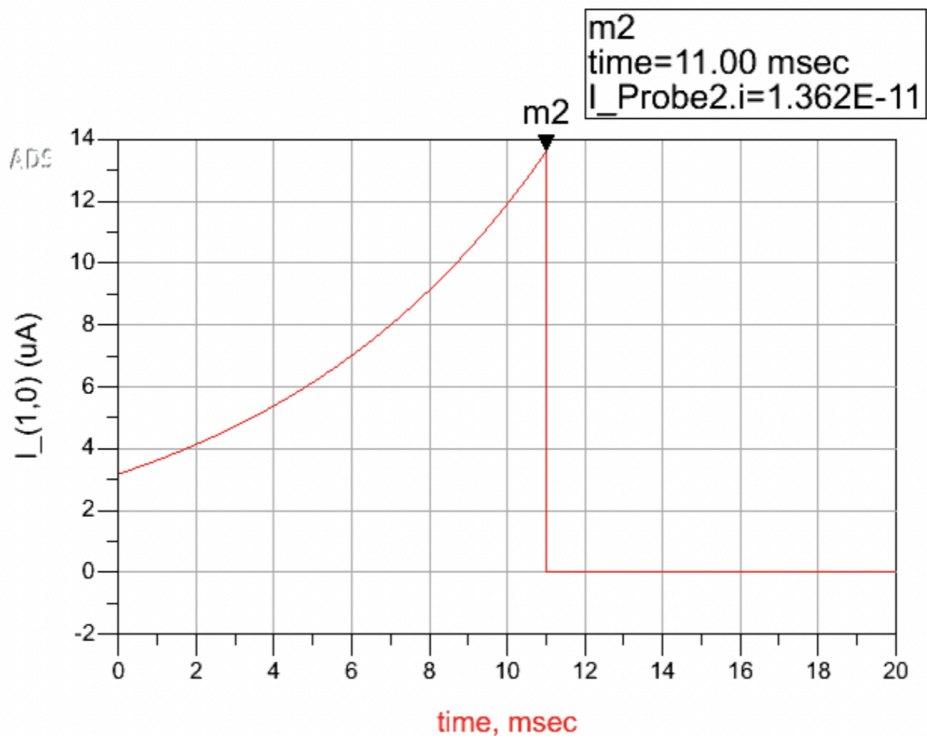


Figure 3. Current through cell(1,0)

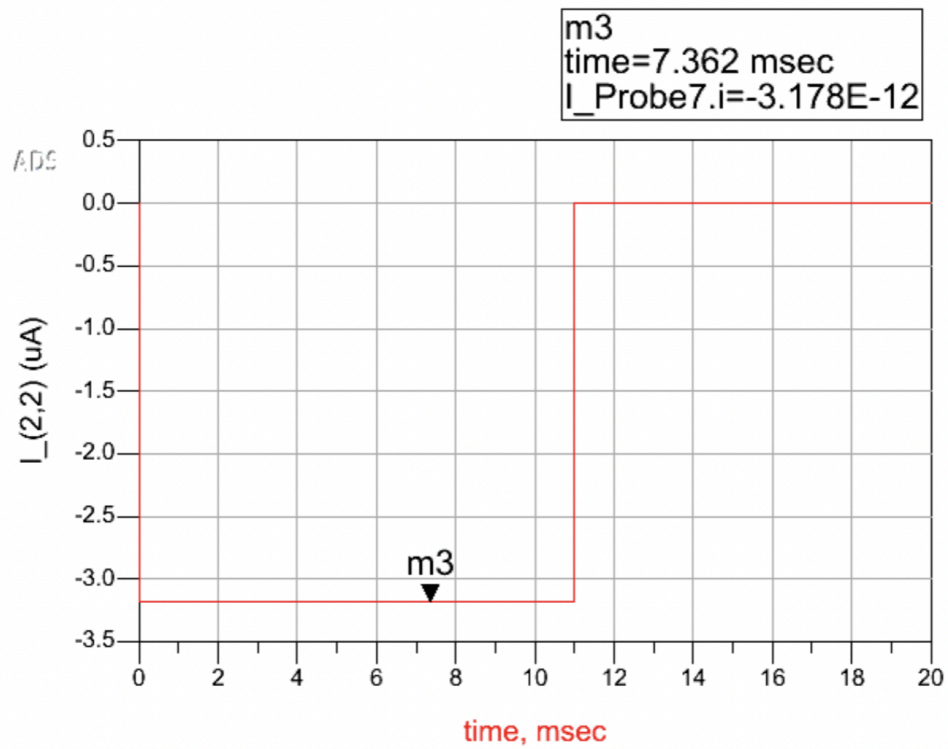


Figure 4. Current through cell(2,2)

For the V/3 Scheme, we supply the following BL and WL voltages:

1. WL for selected cell (0,0) = $V = 1 \text{ V}$
2. BL for selected cell (0,0) = $0 = 0 \text{ V}$
3. WL for all other cells = $V/3 = 0.333 \text{ V}$
4. BL for all other cells = $2V/3 = 0.666 \text{ V}$

Hence,

- the selected cell (0,0) sees $V - 0 = V = 1 \text{ V}$ across the TE and BE
- the half-selected cells (1,0), (2,0) see $V - 2V/3 = V/3 = 0.333 \text{ V}$
- the half-selected cells (0,1), (0,2) see $V/3 - 0 = V/3 = 0.333 \text{ V}$
- The unselected cells (rest of the cells) see $V/3 - 2V/3 = -V/3 = -0.333 \text{ V}$

(c) Find the gap in (1,0) cell

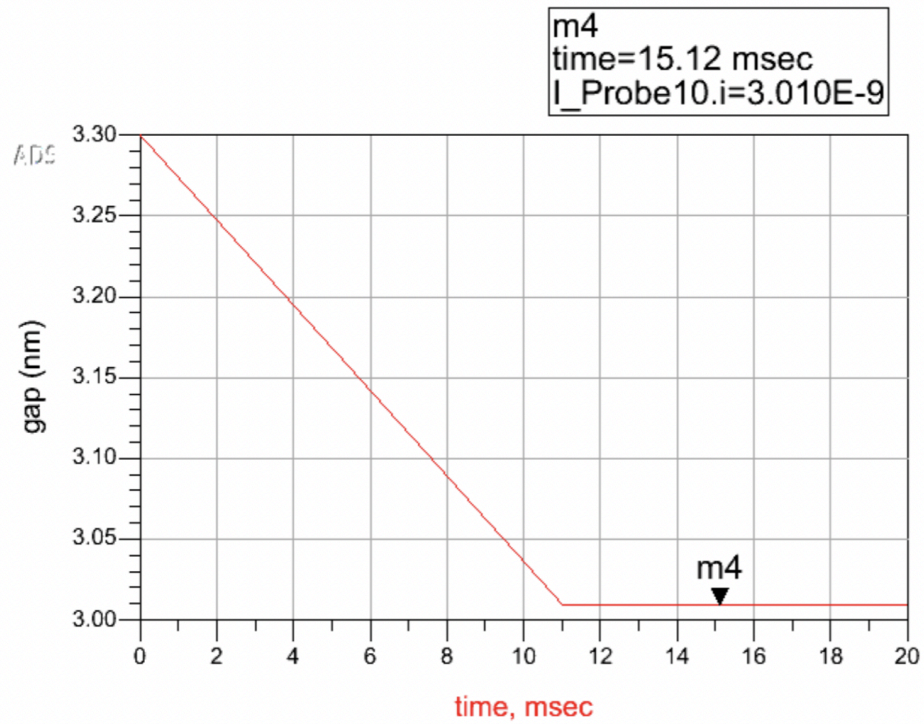


Figure 5. Gap profile for (1,0) cell

From the above gap profile, as long as the pulse is high, the gap reduces, and when the voltage across the TE, BE is 0, the gap value is constant.

Task 2:

- (a) Now create a 3 x 3 1T-1R Crossbar. You can use the same transistor settings we used in the Mid-Sem exam. (For writing, use a 1.2V pulse for 10ms to get an ON conductance of about 0.4nS)

Ans:

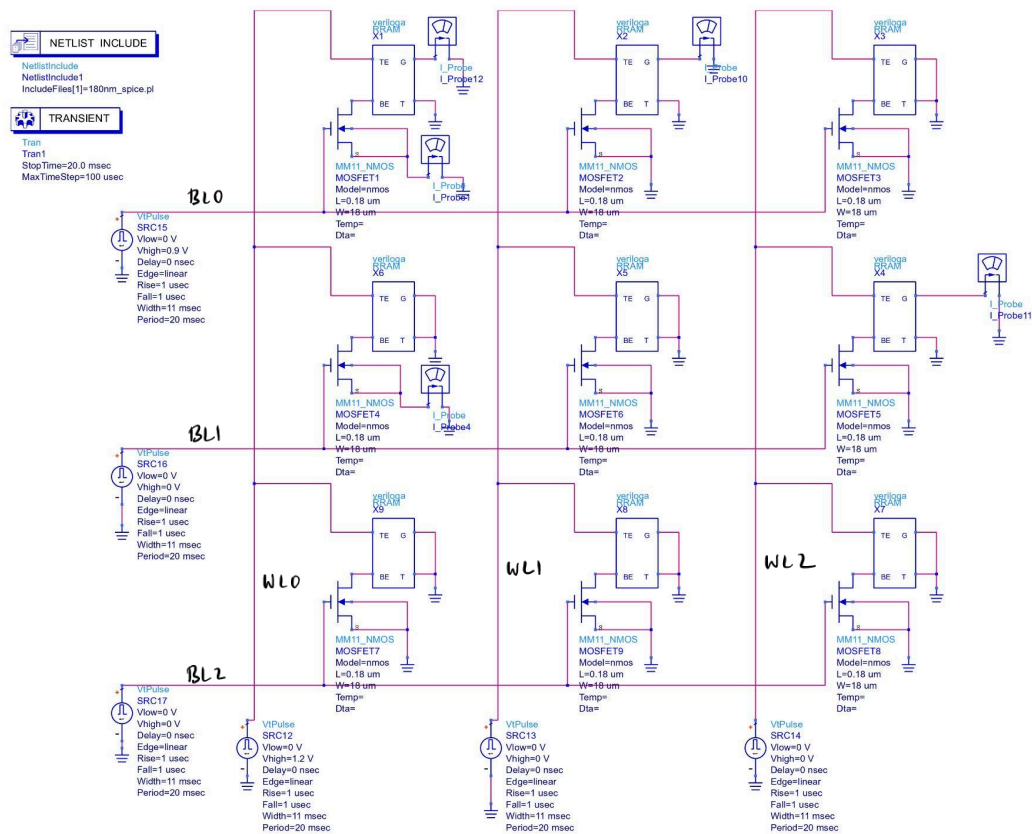


Figure 6. Schematic for 1T1R 3x3 crossbar, assuming $SL = 0$ common source

- (b) Follow the same steps, but now instead of the V/3 scheme, just turn the WL of the 2nd and 3rd rows off. Use the BLs of the 1st row to program (0,0) appropriately

Check above schematic for the WL and BL values

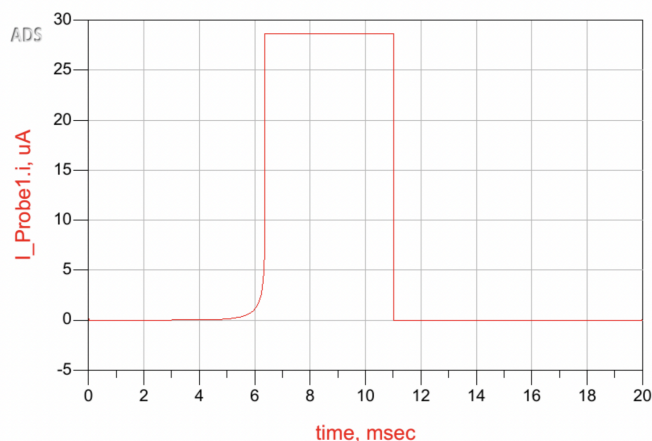


Figure 7. Current through 1T1R for selected Cell

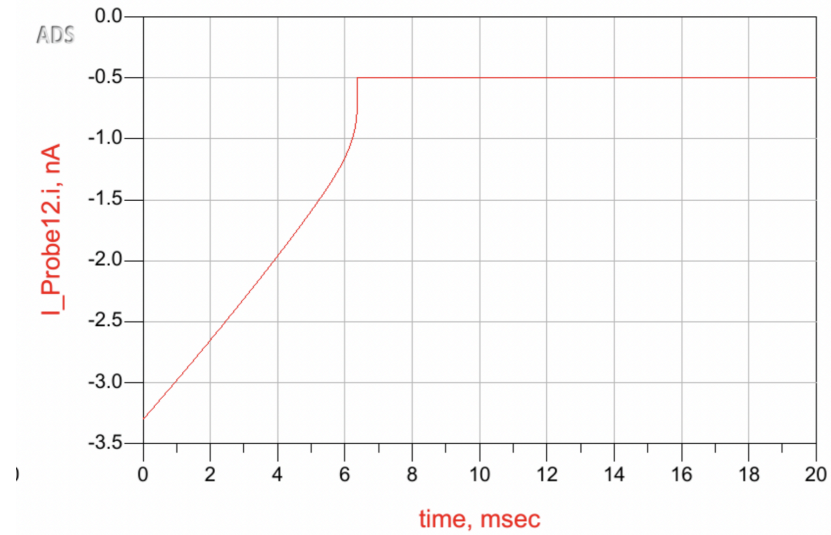


Figure 8. Gap plot for the selected cell, here the gap is reducing significantly, hence the 1T1R is programmed.

(c) Now read the gap in (1,0) cell. Compare it with the previous part

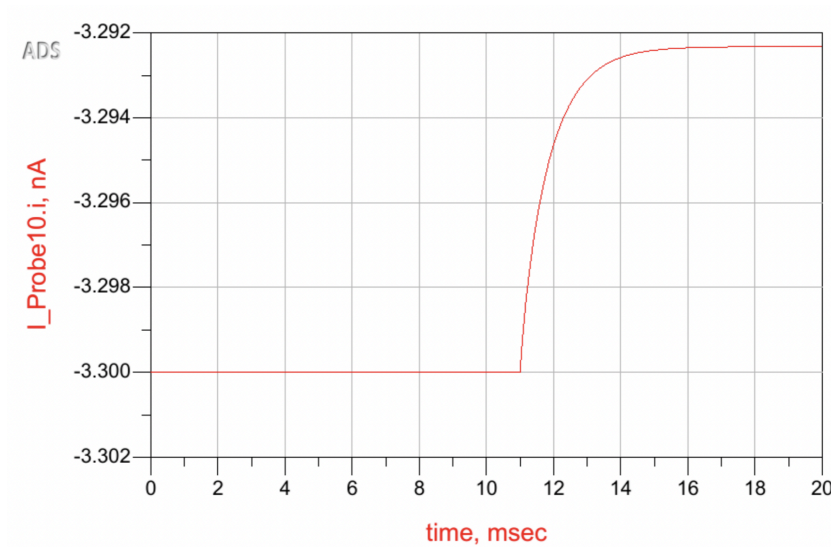
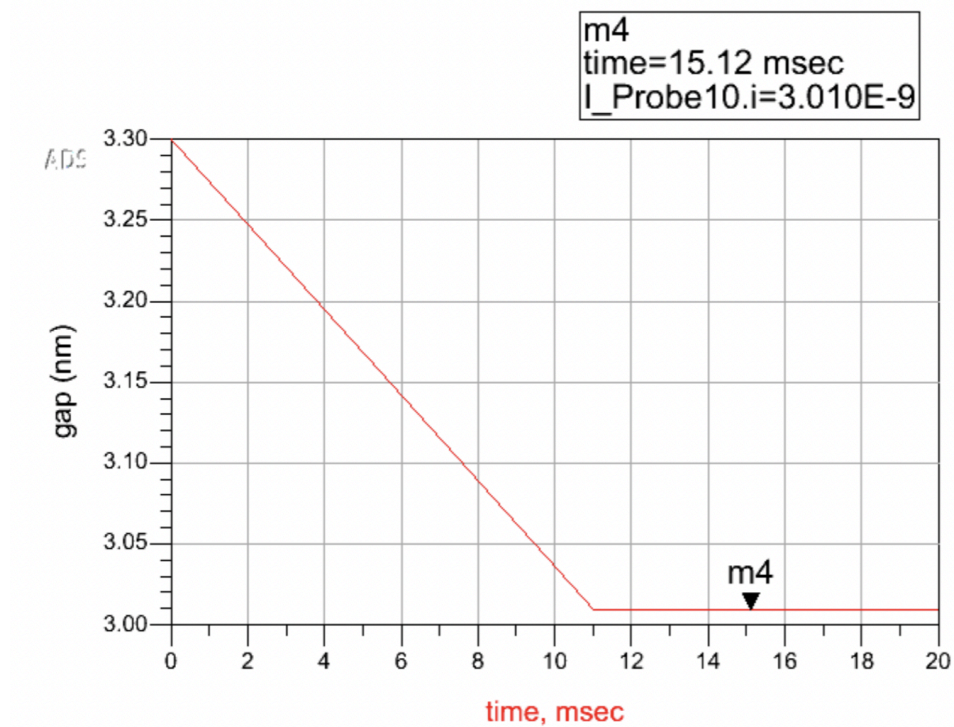


Figure 9. Gap plot for the selected cell, here the gap has not reduced significantly, hence the 1T1R is not programmed.

Comparing this with Task 1's gap plot, which is:



We see that the gap in 1T1R half-selected cell has not reduced significantly:

3.3 nm → 3.292 nm

Whereas, the gap in RRAM half-selected cell has reduced from:

3.3 nm → 3.01 nm

The reason for this is the reduction is sneak current in the 1T1R half-selected cells, which significantly reduces the power usage and also improves the memory window for the half-selected cell.

Task 3:

(a) Now instead of writing cell-by-cell. Write the following weights row-by-row. Report the number of pulse-cycles taken to program this.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Ans:

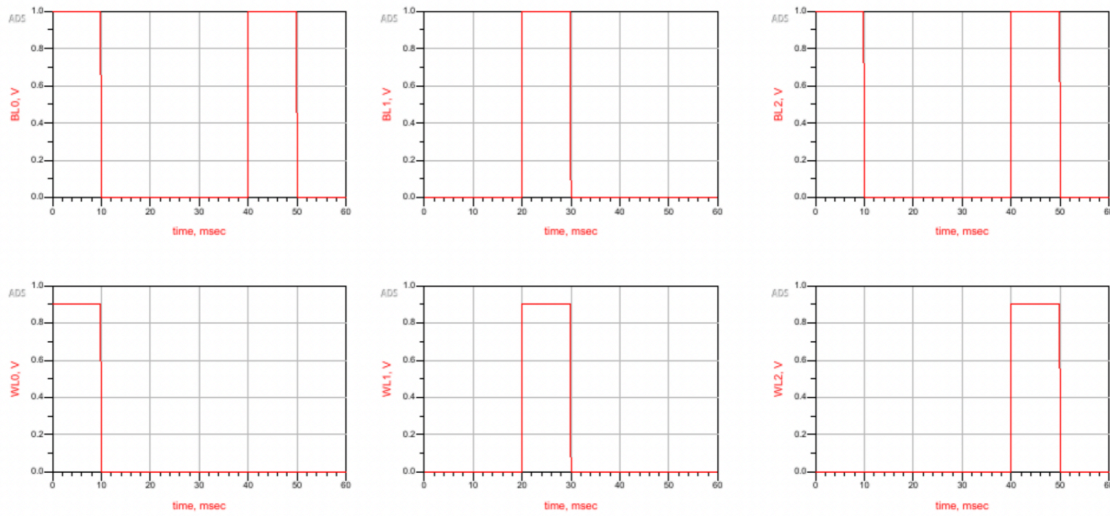


Figure 10. Input WLs and BLs to the cells, assuming $SL = 0$ common source

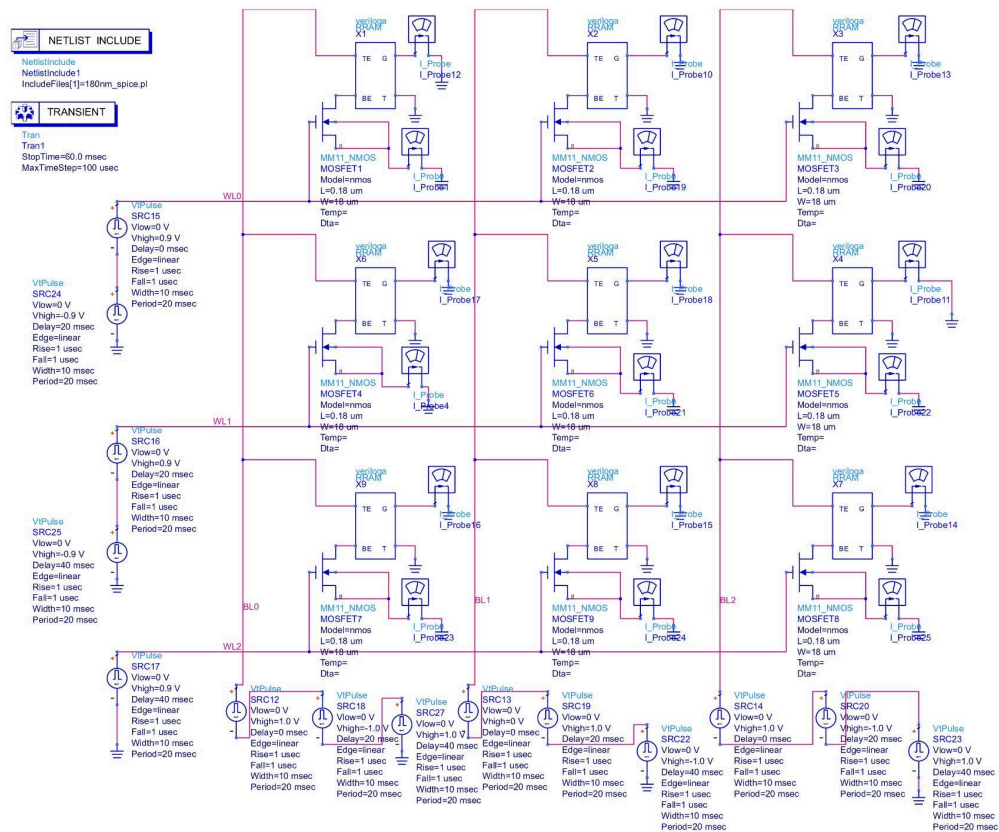


Figure 11. Schematic for storing synaptic weights in the crossbar array

The operation for storing the given synaptic weights is done in 3 cycles as seen from the inputs provided.

(b) Find the gaps of all the cells and report them

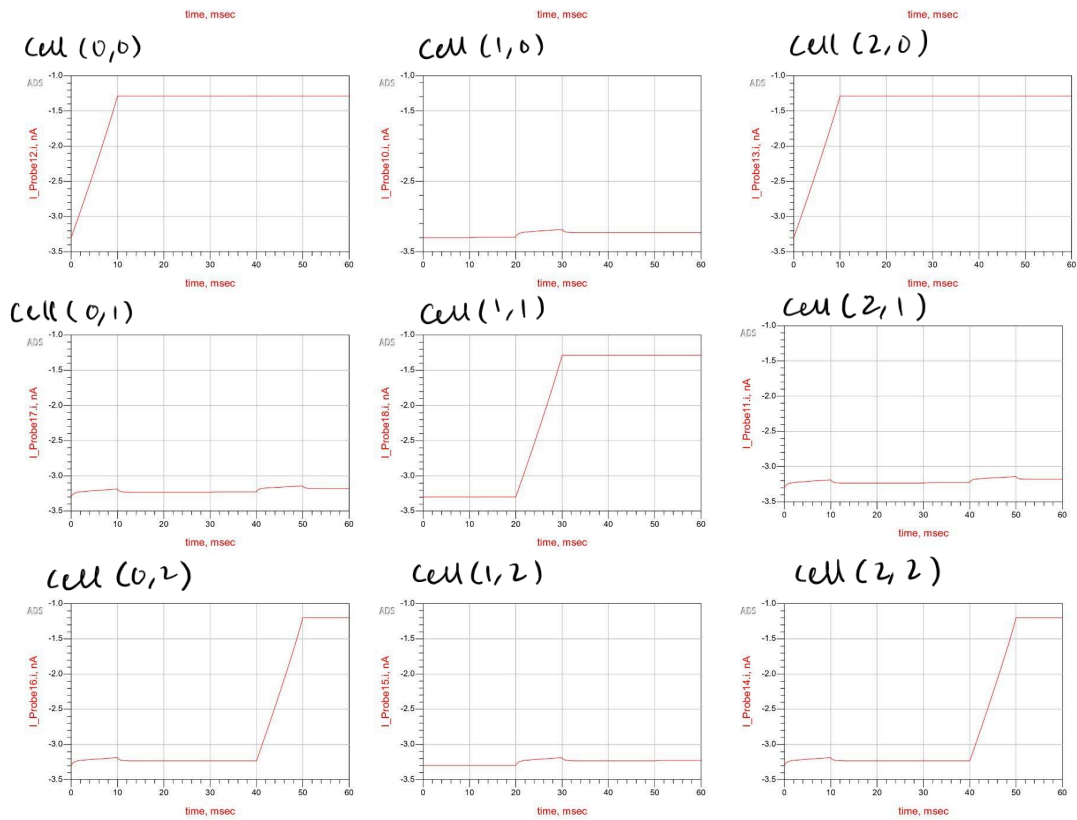


Figure 12. Gap plots for all the cells in the corresponding 3x3 ordered matrix (scaled)

It is clear from the above plots that the crossbar circuit above functions as a memory array for storing a 3x3 matrix.

There is a **significant reduction** in the gaps for the **programmed (selected)** 1T1R cells and in the case of **unselected** and **half-selected** the gaps are almost the **constant** with time.

change in gap $\rightarrow$ programmed
 constant gap $\rightarrow$ not programmed } principle of storing logic in 1T-1R crossbar.